

Given

$$1) a. k = 0.292 \left(\frac{W}{m \cdot K} \right)$$

$$(T = 273 + 3x + 4y + 5z) \text{ (K)}$$

Work
 ↳ 3D
 ↳ isotropic
 $q = -k \nabla T$

$$\vec{q} = -(0.292 \frac{W}{m \cdot K}) \cdot (\vec{e}_x \frac{\partial}{\partial x} [T] + \vec{e}_y \frac{\partial}{\partial y} [T] + \vec{e}_z \frac{\partial}{\partial z} [T])$$

unknown
 $\vec{q} = ?$

$$Eq \quad q = \frac{\dot{Q}}{A} = \left[\frac{J}{s m^2} \right] = \left[\frac{W}{m^2} \right]$$

$$\vec{q} = -k \nabla T$$

$$= -k \cdot \left(\sum_{i=1}^3 \vec{e}_i \frac{\partial T}{\partial x_i} \right) [T(x_i)]$$

Break

$\frac{\partial}{\partial x} [T]$	$\frac{\partial}{\partial y} [T]$	$\frac{\partial}{\partial z} [T]$
$= \frac{\partial}{\partial x} [273 + 3x + 4y + 5z] k$	$= \frac{\partial}{\partial y} [273 + 3x + 4y + 5z]$	$= \frac{\partial}{\partial z} [273 + 3x + 4y + 5z]$
$= 3 \text{ K/m}$	$= 4 \text{ K/m}$	$= 5 \text{ K/m}$

Back

$$\vec{q} = -(0.292 \frac{W}{m \cdot K}) \cdot (3\vec{e}_x + 4\vec{e}_y + 5\vec{e}_z) \text{ K/m}$$

$$= (-0.876\vec{e}_x - 1.168\vec{e}_y - 1.46\vec{e}_z) \text{ W/m}^2 \leftarrow [W/m^2]$$

b. Given

$$\vec{k} = \begin{bmatrix} 10 & 5 & 0.38 \\ 5 & 10 & 5 \\ 0.38 & 5 & 10 \end{bmatrix} \left(\frac{W}{m \cdot K} \right) \rightarrow \text{tensor}$$

$$T = 273 + 3x + 4y + 5z \text{ (K)}$$

↳ 3D

unknown
 $\vec{q} = ?$

$$\vec{q} = -\vec{k} \nabla T$$

$$= -\vec{k} \left(\sum_{i=1}^3 \vec{e}_i \frac{\partial T}{\partial x_i} \right) [T(x_i)]$$

Work
 ① $\nabla T = \frac{\partial T}{\partial x} \vec{e}_x + \frac{\partial T}{\partial y} \vec{e}_y + \frac{\partial T}{\partial z} \vec{e}_z$
 ② $\nabla T = 3\vec{e}_x + 4\vec{e}_y + 5\vec{e}_z$

$$\vec{q} = -\vec{k} \nabla T$$

$$= \begin{bmatrix} 10 & 5 & 0.38 \\ 5 & 10 & 5 \\ 0.38 & 5 & 10 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

$3 \times 3 \quad \quad \quad 3 \times 1$

$$= [51.9, 80, 71.14]$$

$$\Rightarrow \vec{q} = -51.9\vec{e}_x - 80\vec{e}_y - 71.14\vec{e}_z$$

2) Given

$$\begin{aligned} A &= 100 \text{ cm}^2 \\ Y &= 0.5 \text{ cm} \\ Q &= 3 \text{ W} \\ T_0 &= 25^\circ \text{C} \\ T_1 &= 30^\circ \text{C} \end{aligned}$$

steady state

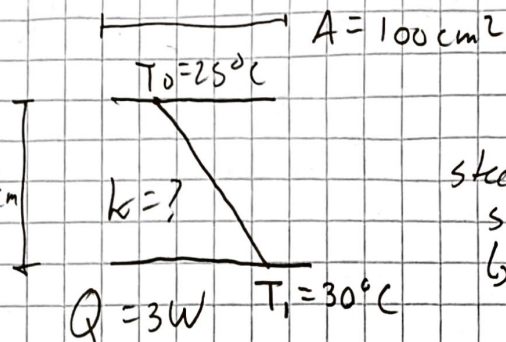
Unknown
 $k = ?$

$$\frac{Q}{A} = -k \frac{\Delta T}{Y} \quad Y = 0.5 \text{ cm}$$

$$k = \left[\frac{\text{J}}{\text{m} \cdot \text{s} \cdot \text{K}} \right]$$

$$k = \left[\frac{\text{W}}{\text{m} \cdot \text{K}} \right]$$

positive value



steady
state
 $\hookrightarrow T \gg 0$

Work $\frac{Q}{A} = -k \frac{\Delta T}{Y}$

$$\hookrightarrow -\frac{Q \cdot Y}{A \Delta T} = k$$

$$k = \frac{-3 \text{ W} \cdot 0.5 \text{ cm}}{100 \text{ cm}^2 \cdot (25^\circ \text{C} - 30^\circ \text{C})} \quad \text{good}$$

$\rightarrow k \Delta T$

$$k = -0.003 \frac{\text{W}}{\text{cm} \cdot \text{K}}$$

✓